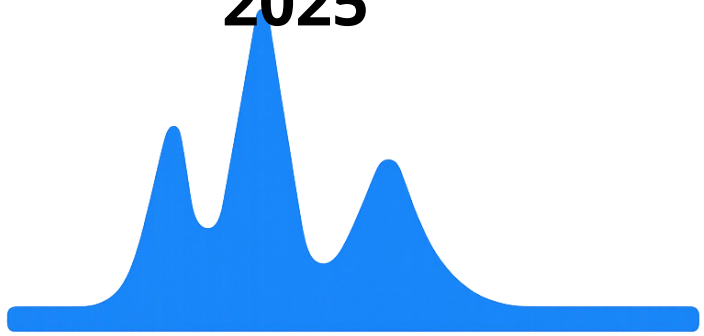
CSN 5 - GRANT GIOVANI RICERCATORI 2025



PHASE-*g*

Perovskite-**H**ybrid **A**rchitecture for **S**pectroscopy and **E**nergy-Counting of **γ**-Rays

Proposer:
I. Margotti

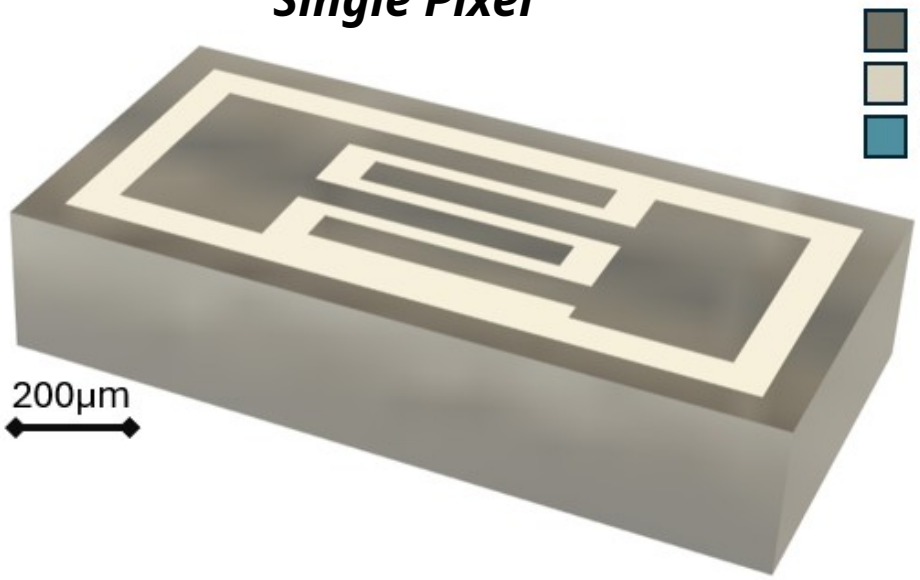
Members: **Dr. A. Ciavatti, Dr. I. Fratelli, Dr. S.M. Carturan, Dr. C. Sbarra, Prof. B. Fraboni**

PEROVSKITE-Based γ-DETECTORS

A Proof of Concept towards printable, high-performance devices

Gamma SPECTROSCOPY (GS)

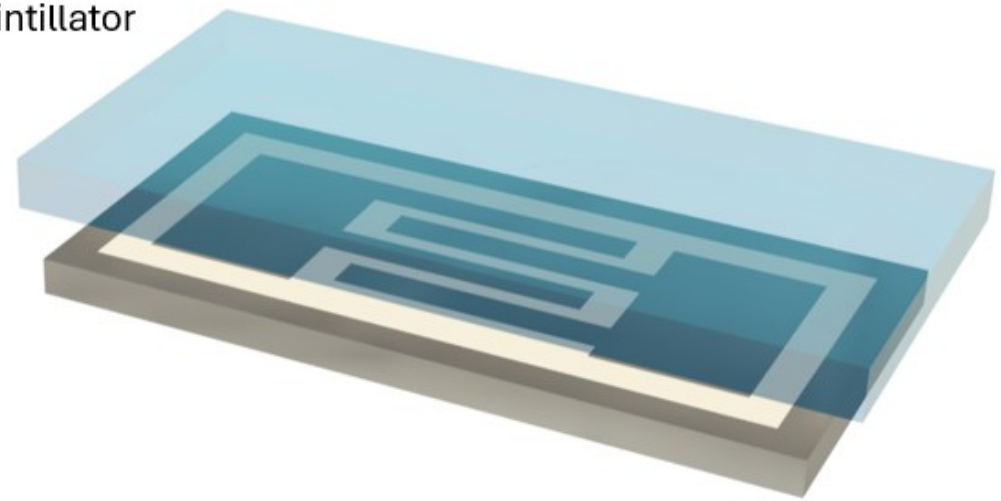
Single Pixel



Metal contacts
2D perovskite
Organic Scintillator

PHOTON COUNTING Dev (PCD)

Single Pixel



PEROVSKITE-Based γ-DETECTORS

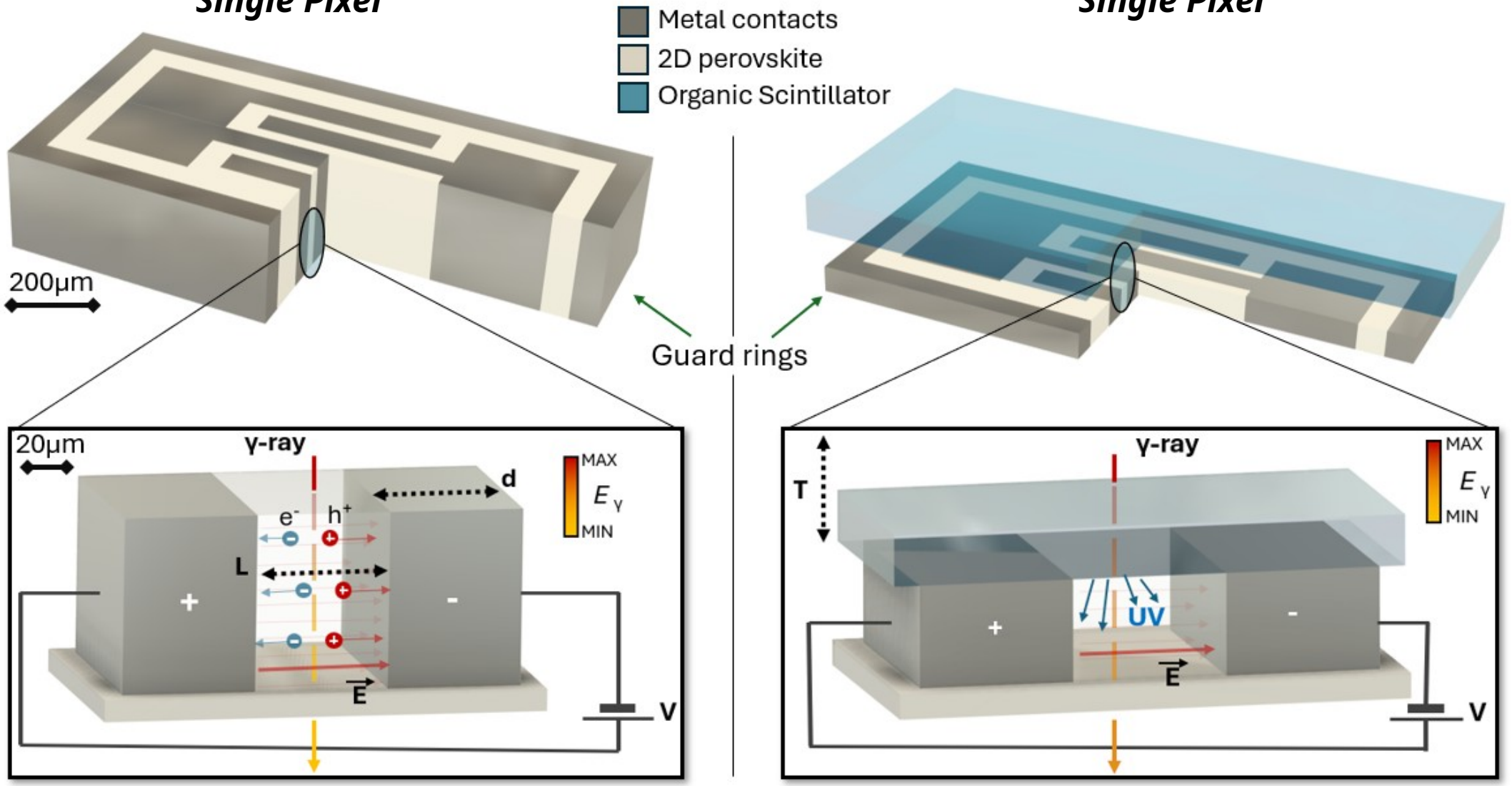
A Proof of Concept towards printable, high-performance devices

Gamma SPECTROSCOPY (GS)

Single Pixel

PHOTON COUNTING Dev (PCD)

Single Pixel



PEROVSKITE-Based γ-DETECTORS

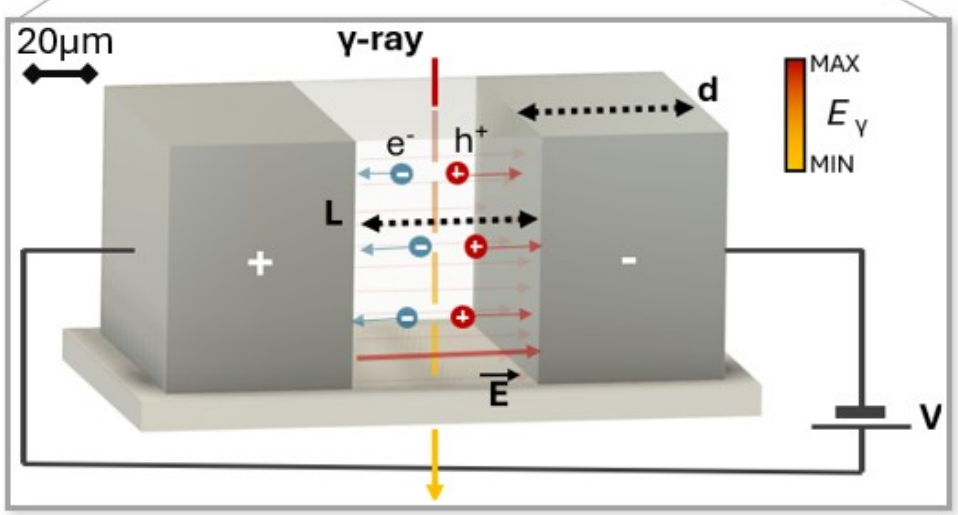
A Proof of Concept towards printable, high-performance devices

Gamma SPECTROSCOPY (GS)

Single Pixel

Detector type: **DIRECT**

- Area: **A < 1mm²**
- Target aspect ratio: **(10:1)**
- Key feature: ***high thickness for enhanced interaction volume***

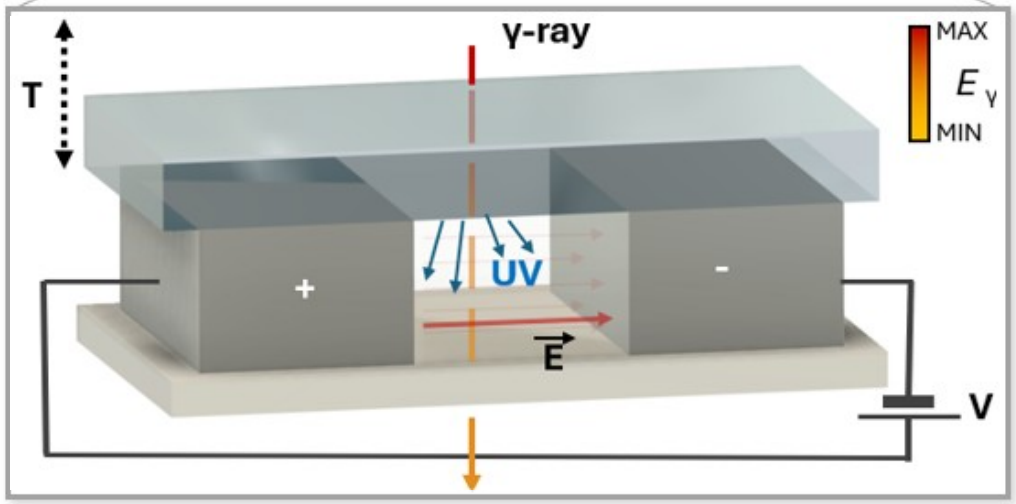


PHOTON COUNTING Dev (PCD)

Single Pixel

Detector type: **INDIRECT/HYBRID**

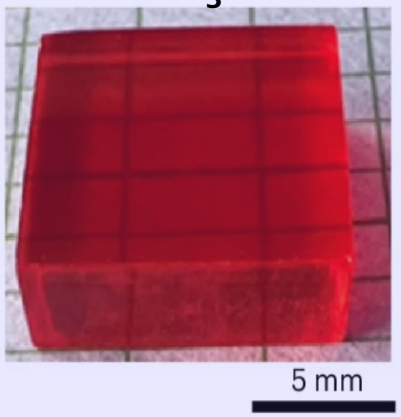
- Area: **A < 1mm²**
- Target aspect ratio: **(1:1)**
- Key feature: **organic-plastic scintillator on top *for UV emission***

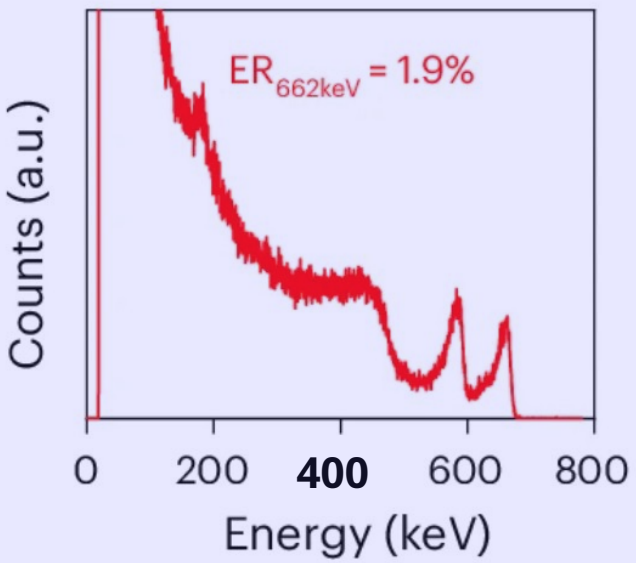


SINGLE CRYSTALS VS THIN FILMS

State of the art: coupling bulk performance with handling features

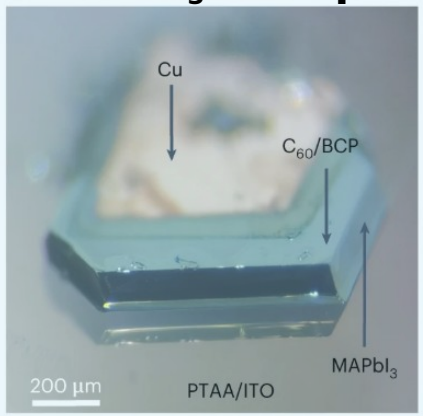
$FAPbBr_3$ – 3mm

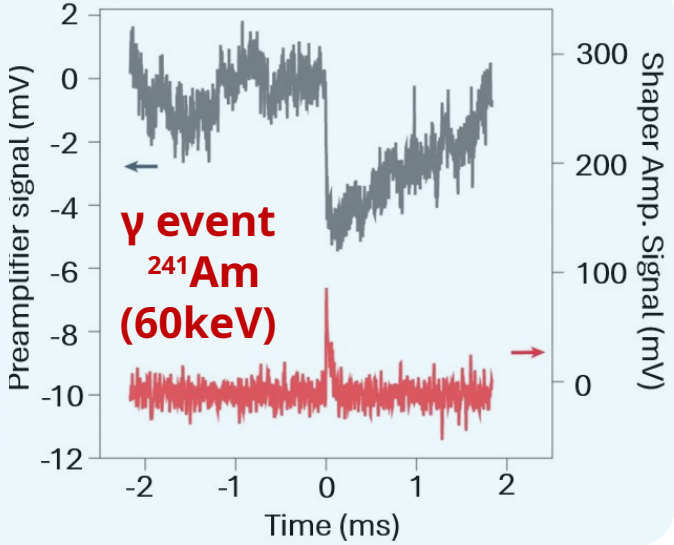




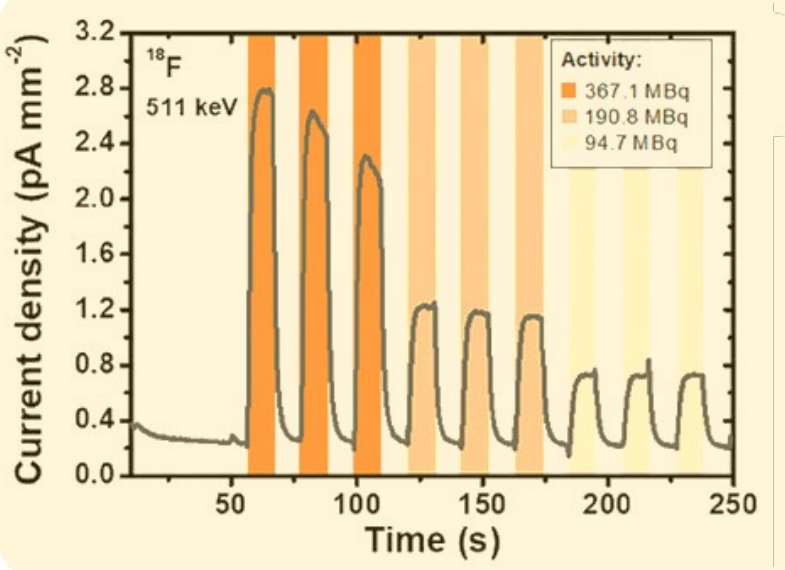
Li, M., Wang, S., Wood, A. *et al.*
Nat. Mater. (2025)

$MAPbI_3$ - 110 µm

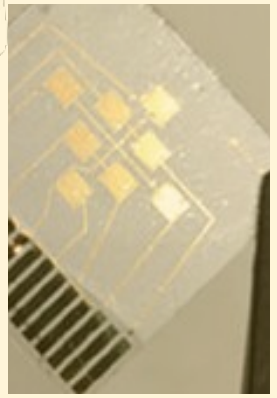




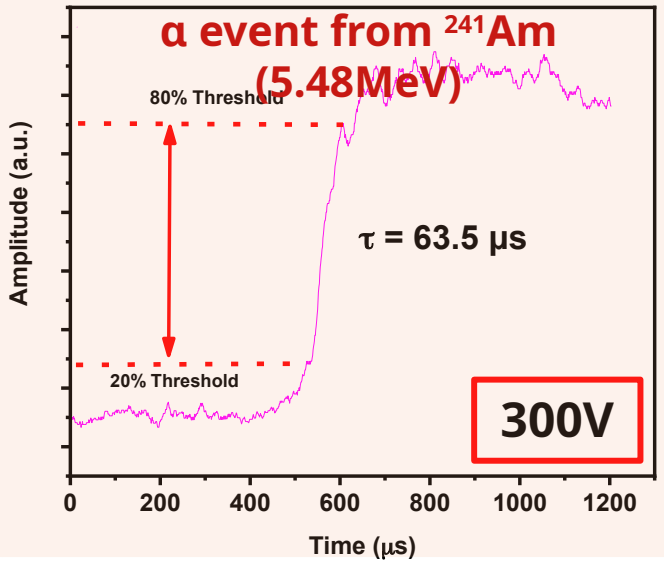
Sakhatskyi, K. *et al.*
Nat. Photon. (2023)



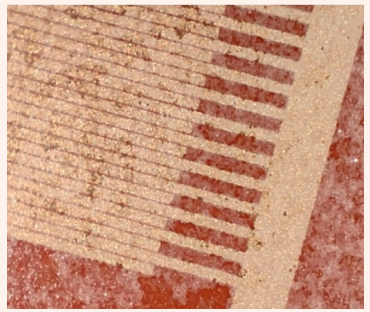
PEA_2PbBr_4



Basiricò, L. *et al.*
Adv. Health. Mat.
(2024)

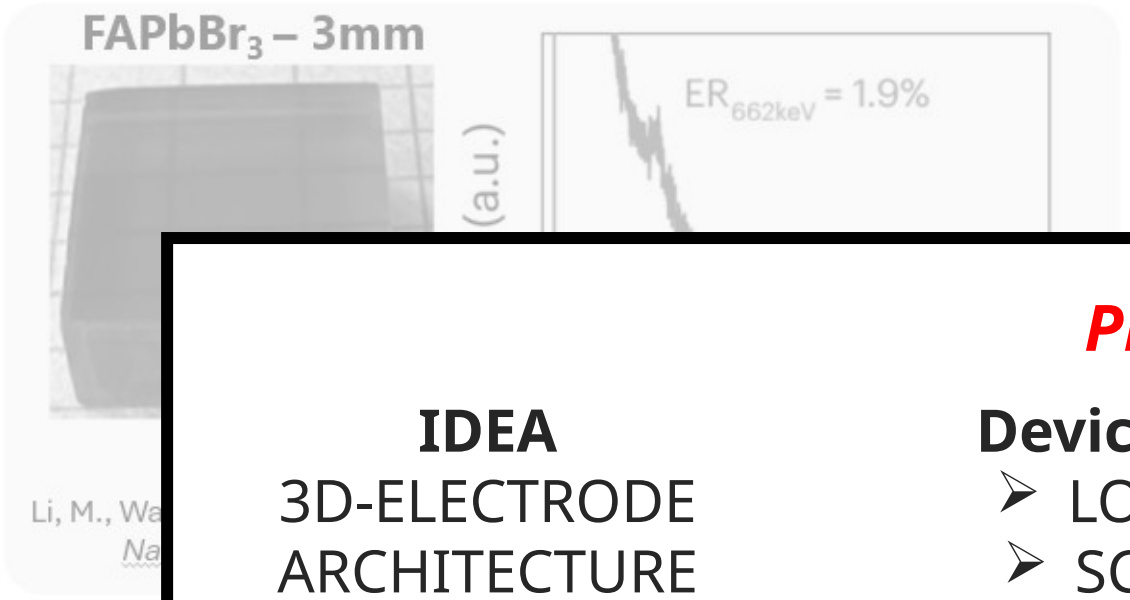


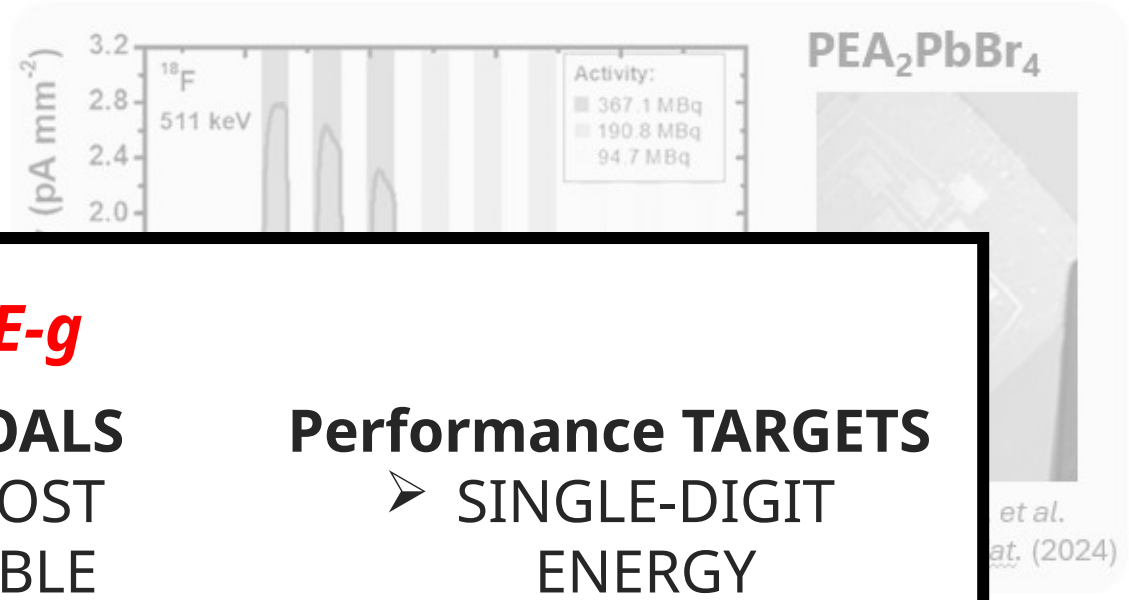
PEA_2PbBr_4



Margotti, L. *et al.*
IEEE NSC, MIC, RTSD
2024

SINGLE CRYSTALS VS THIN FILMS





PHASE-g

IDEA

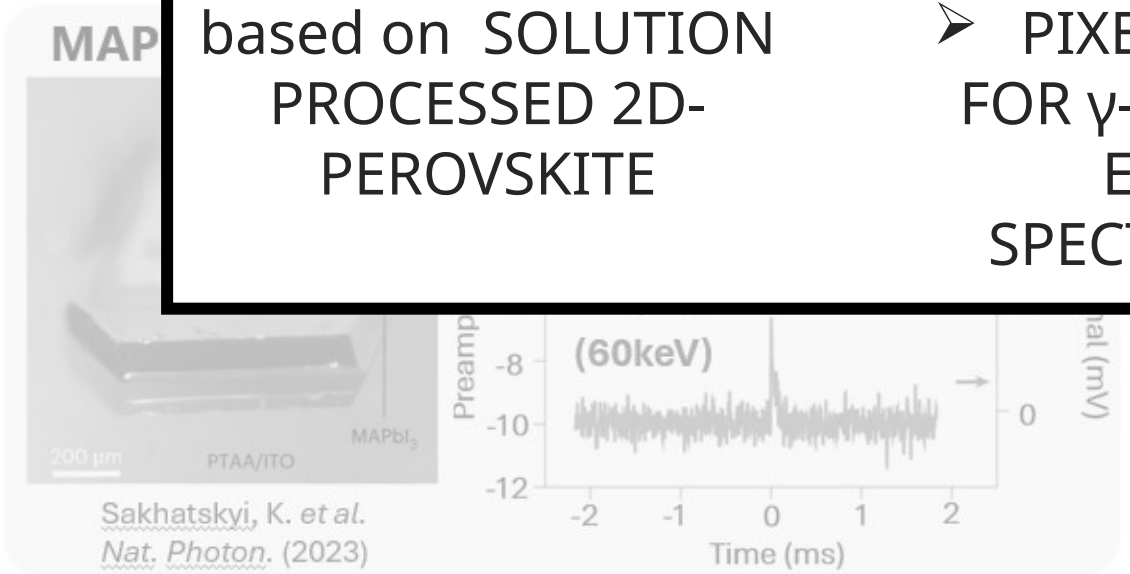
3D-ELECTRODE ARCHITECTURE based on SOLUTION PROCESSED 2D-PEROVSKITE

Device GOALS

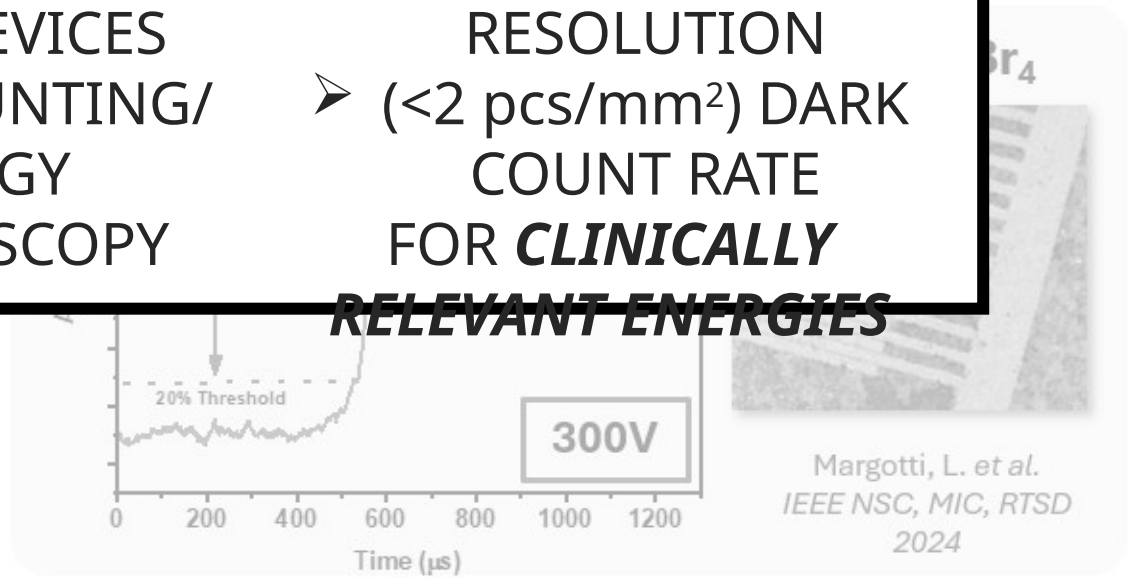
- LOW COST
- SCALABLE
- PIXEL DEVICES FOR γ-COUNTING/ ENERGY SPECTROSCOPY

Performance TARGETS

- SINGLE-DIGIT ENERGY RESOLUTION
- (<2 pcs/mm²) DARK COUNT RATE FOR ***CLINICALLY RELEVANT ENERGIES***

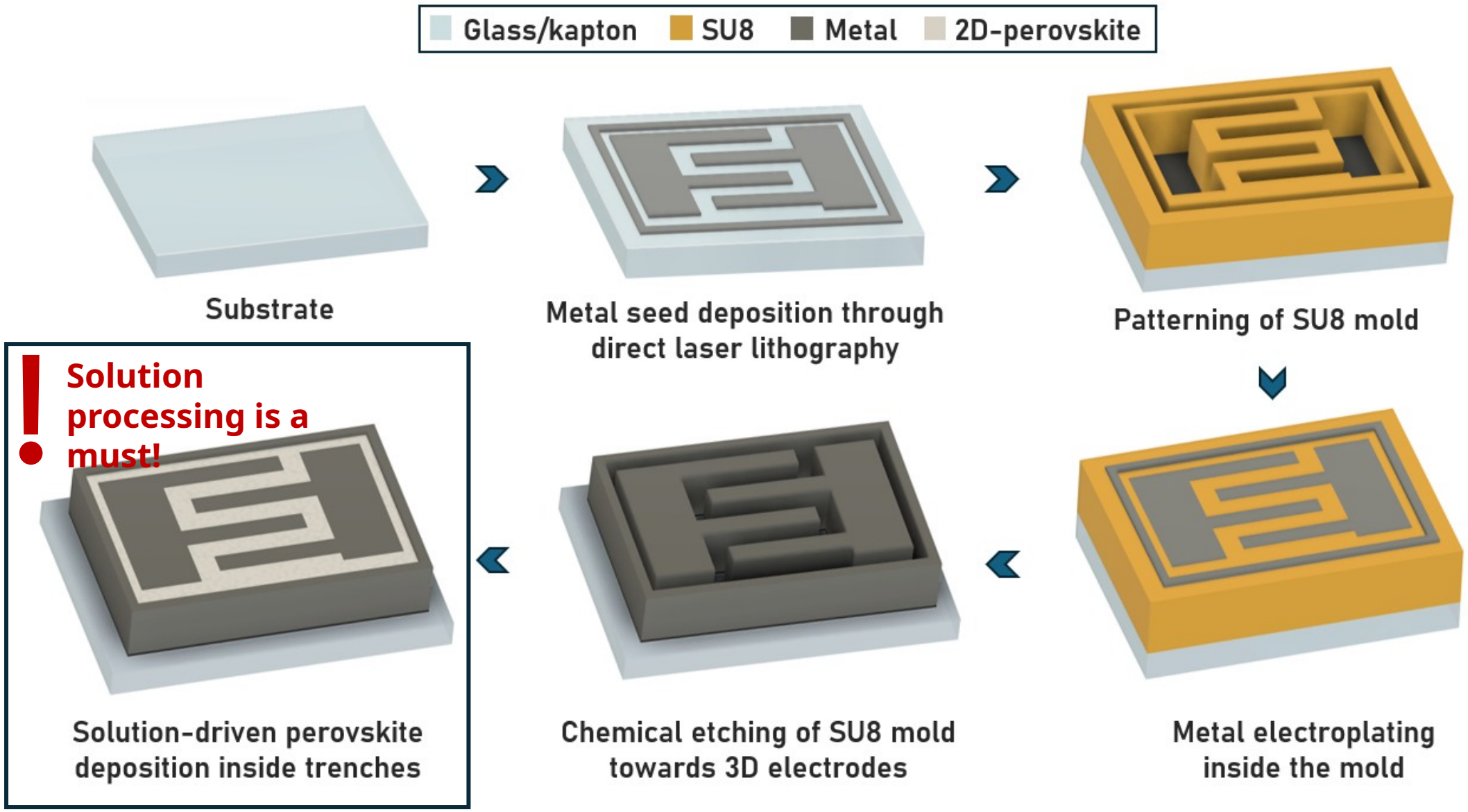


Sakhatskyi, K. *et al.* *Nat. Photon.* (2023)



Margotti, L. *et al.* *IEEE NSC, MIC, RTSD 2024*

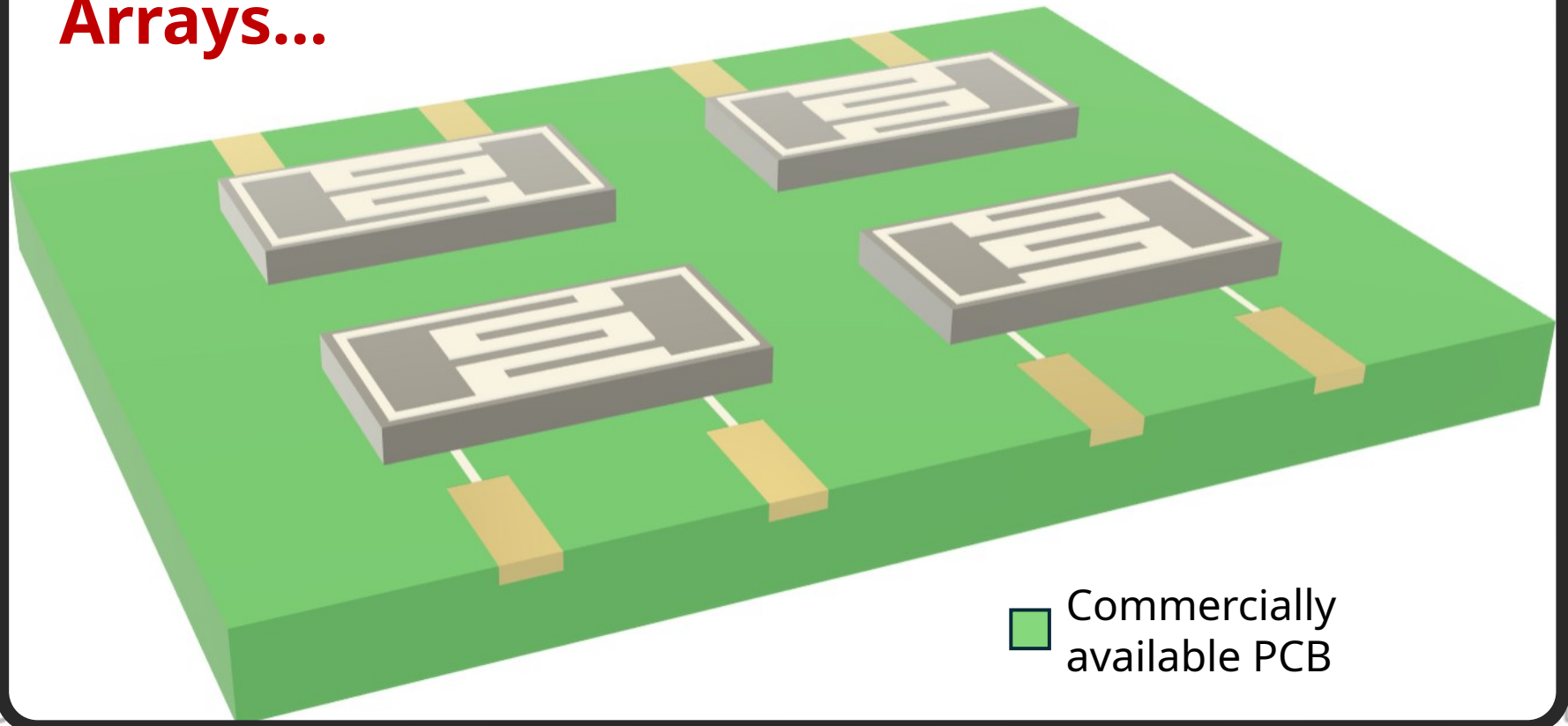
PEROVSKITE-Based DETECTORS



PEROVSKITE-Based DETECTORS

Glass/kapton SU8 Metal 2D-perovskite

Envisioning PCB-printed Multielectrode Arrays…



Commercially available PCB

deposition inside trenches

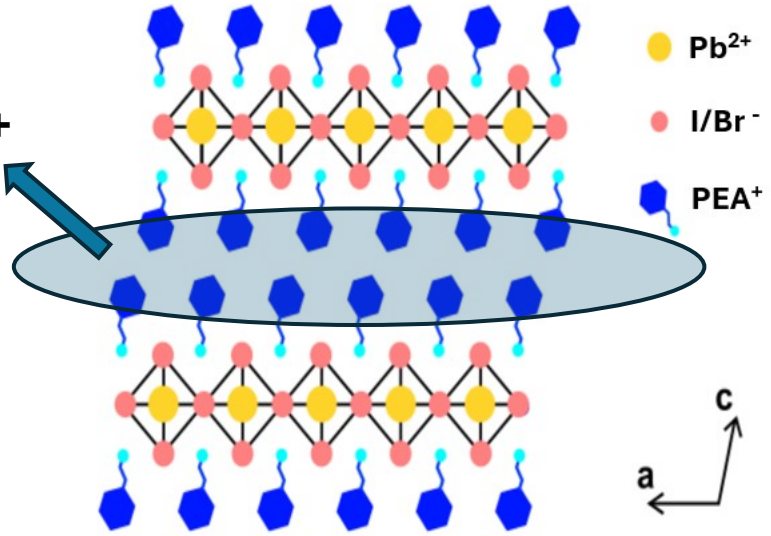
towards 3D electrodes

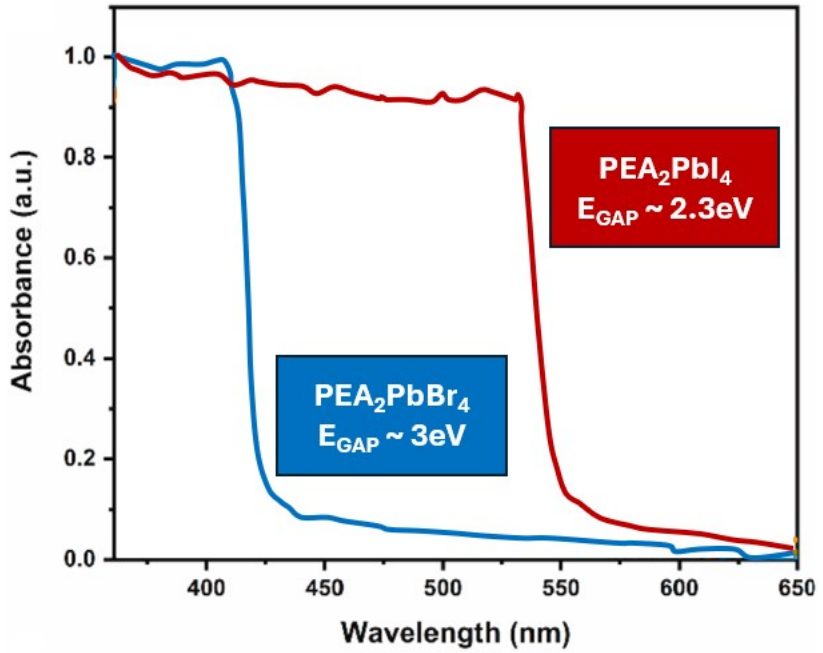
inside the mold

2D PEROVSKITES: $PEA_2Pb(Br/I)_4$

Featuring highly resistive, more stable structures

Quantum + dielectric confinements

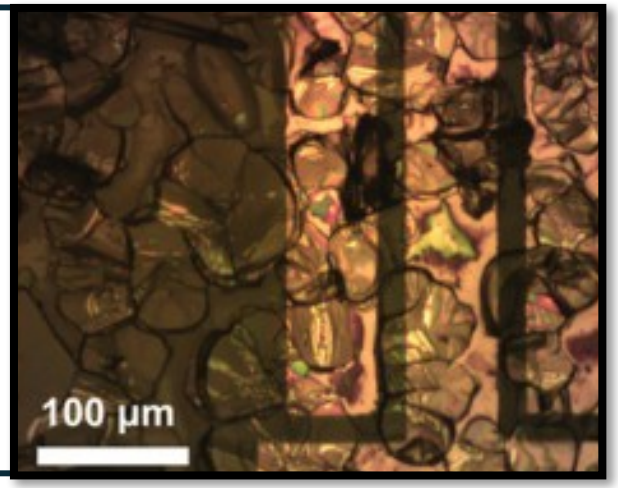




Features of interest compared to 3D perovskites

✅ Larger bandgap → Fewer thermally excited carriers
✅ Lower trap density → Reduced trap-assisted leakage
✅ Dielectric confinement → Less unintentional carrier injection
✅ Reduced ion migration → Limited degradation

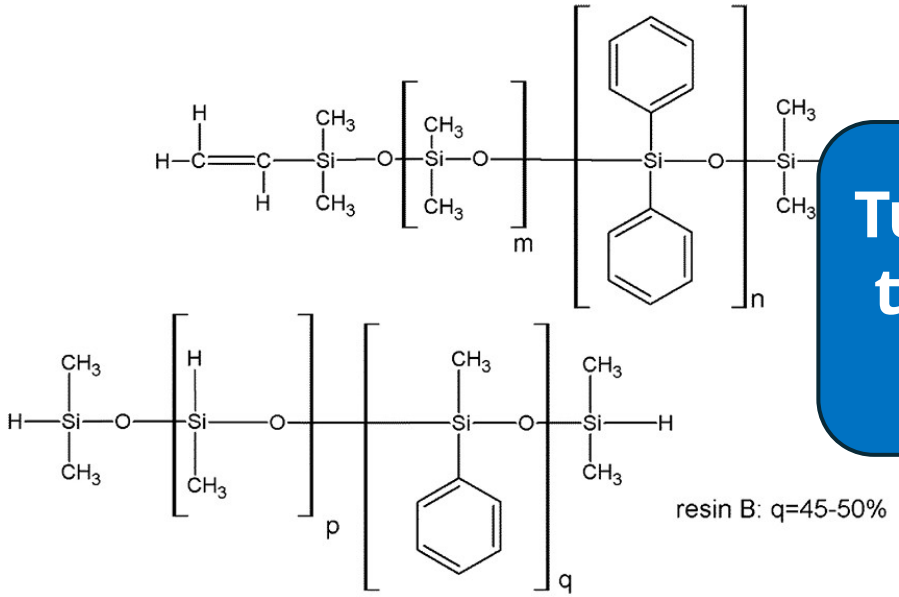
➔ **Lower dark current (10^{-13}A @ 1kV/mm)**
➔ **Higher stability**



ORGANIC-PLASTIC SCINTILLATOR

Materials and Coupling

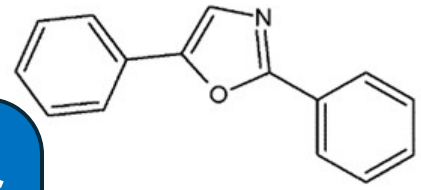
PMPS :polydimethyl-co-diphenylsiloxane scintillators



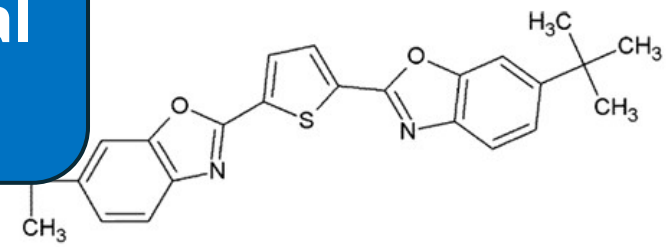
resin B: q=45-50%

Tunable properties through chemical optimization!

Phenyl groups /dyes: examples



2,5-diphenyloxazole (PPO)



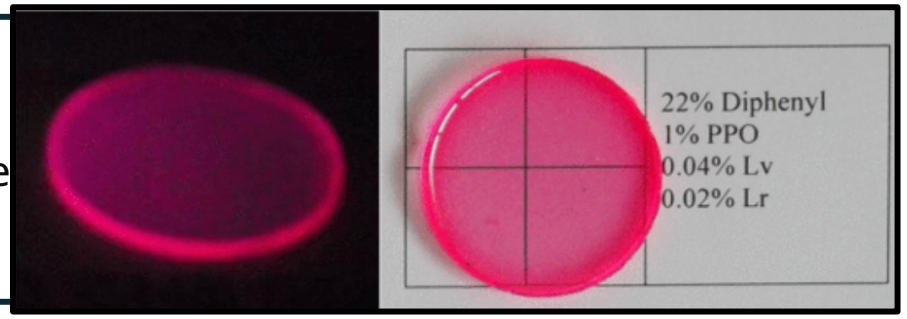
2,5-Bis (5-tert-butyl-benzoxazol-2-yl)thiophene (BBOT)

Goals:

✅Peak brilliance in the UV range (320-380nm)

✅Light yield under γ-sources at RT (60% rel. EJ212 unde ^{137}Cs γ-rays, i.e. 5/7x10^{3} ph/MeV))

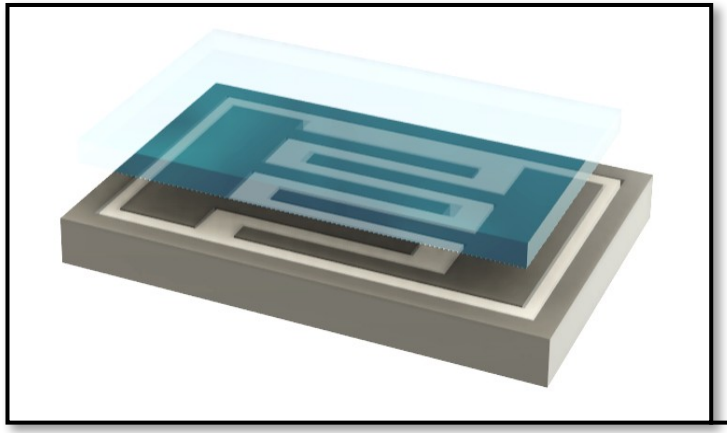
✅Precise optical coupling with 2D perovskites



M. Dalla Palma *et al.* , 2024,
IEEE Transactions on Nuclear Science

OPTIMIZATION OF GEOMETRICAL PARAMETERS

GEANT4 for feedback-driven analyses



PARAMETERS OF INTEREST

- Electric field inside the trenches
- Structural features (thickness, width, area)
- Optical coupling (emission and absorption matching)

FEEDBACK-DRIVEN SIMULATIONS will provide optimal combination of parameters according to fabrication and desired outcomes.

GEANT4 A SIMULATION TOOLKIT

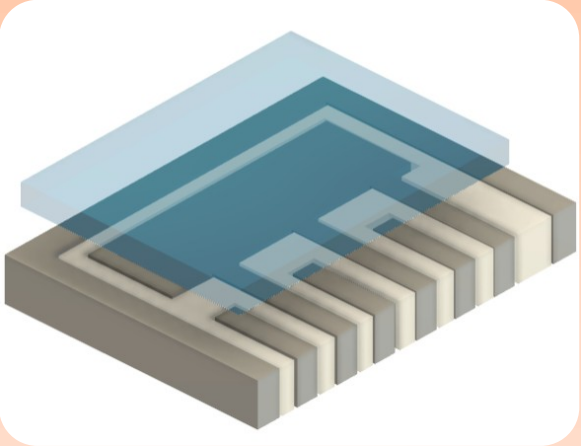
Ansys MAXWELL

Required material physical parameters (density, refractive index, charge carriers mobility, $\mu\tau$ product, absorption/transmission coefficients, light yield) have been assessed in previous works or will be determined through dedicated **tests onsite.**

CORE CHARACTERIZATION

Device-level assessments of GS and PCD

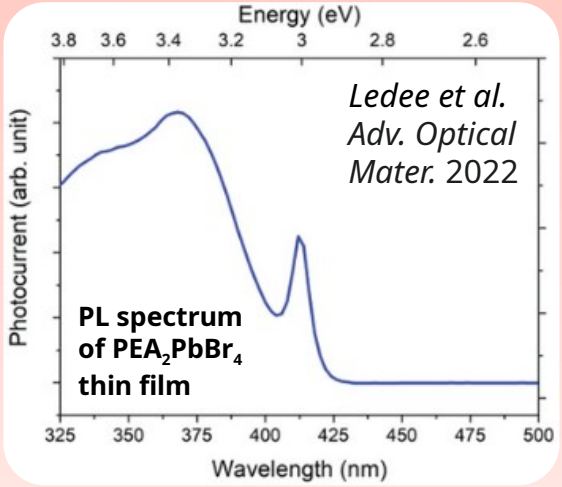
1. STRUCTURAL ANALYSIS



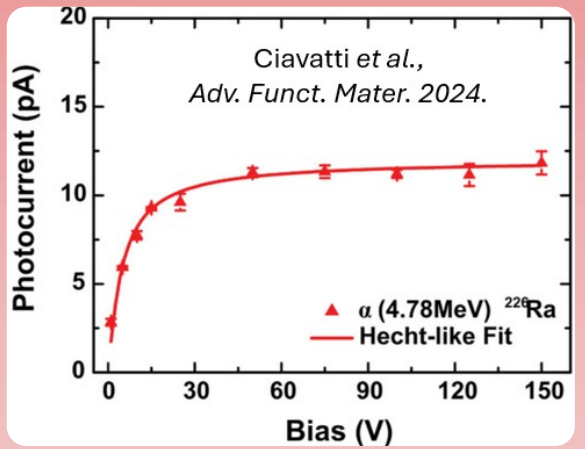
- Cross-section (SEM) for **perovskite filling**
- Surface topography (AFM / 3D profilometry) for **excess material.**

2. BASIC PERFORMANCE ASSESSMENTS

- ***J-V*** and ***C-V*** curves, *PL* and *Abs* spectra
- ***Pulsed analyses***
- Scintillator **photon yield** and **spectral distribution.**



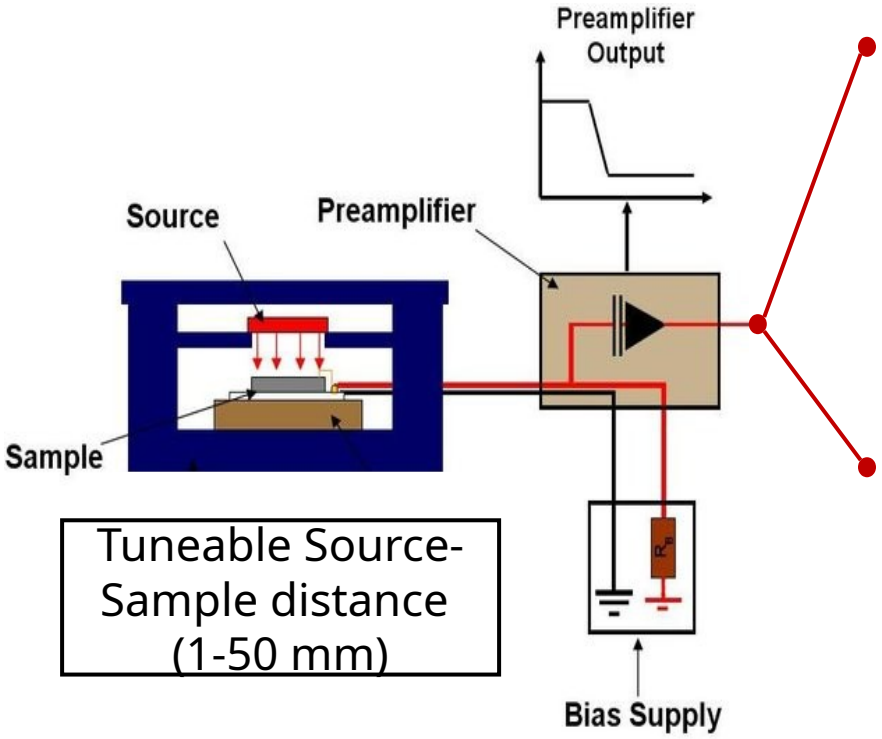
3. RADIATION DETECTION



- **X-rays**: sensitivity, time response, dose-rate linearity, LOD, radiation hardness.
- **α-particles**: single pulse detection, μτ product.

TESTING under GAMMA SOURCES

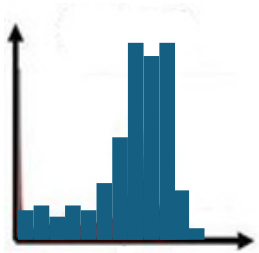
Performance assessments



Tuneable Source-Sample distance (1-50 mm)

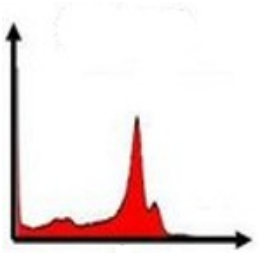
PC: FAST CSA (0.16 fC ENC) + 3.2 GS/s DIGITIZER for nanosecond-scale pulse acquisition.

- **Single-photon detection capability**
- Sensitivity
- Count-rate stability
- Impact of *hybrid contribution*



GS: *low-noise CSA + shaper (up to 100µs) + Digital MCA*

- Charge Collection Efficiency
- Linearity
- **Energy resolution**



Available Gamma Sources		
LOWEST ENERGIES	**HIGHEST ENERGIES**	**MID-RANGE ENERGIES**
^{241}Am	^{137}Cs - ^{60}Co	^{133}Ba
59.5keV	662keV – 1.17 & 1.33MeV	81, 160, 223, 356 keV

Lines in clinically relevant ranges for SPECT (^{99m}Tc, 140keV)!

APPLICATIONS AND PROSPECTS

Expected impact

New prospects for γ-cameras:

✅ Reduction in **volume** and **cost**
✅ Extension to **printable** and **large area devices**
✅ Real-time monitoring of radiotracers and **spectroscopic dosimetry**

Sinergy with the POR-FESR FORTRESS PROJECT (Fraboni) at IRCSS Reggio Emilia, which is now provided a novel SPECT-CT (source: ^{99}mTc, major γ-peak at 140keV).

Compound detectors for nuclear physics

✅ Tiled arrays for localized γ-radiation mapping
✅ Low-dose γ-monitors

Onboard dosimeters

✅ Low-dose γ-monitors
Potential follow-up for the *ASI-VUS3 IRIS project (INFN-TTlab)* for **Wearable Dosimeters for the Crew Members Onboard the ISS.**

TECHNICAL CONSIDERATIONS

Risk Management and Requests

Possible Issues and Mitigation

- **Empty trenches / excess material**
  - Sonication-assisted deposition / lapping techniques
- **Environmental issues / external light contribution**
  - Deposition of Parylene-C encapsulation layer
- **Poor γ absorption**
  - SJI / Metal Ink deposition for trenches height increased up to 500µm.
- **Interface losses between perovskite and scintillator**
  - Dedicated resin / cement of matched refractive index
- **Delays in electronic instrumentation delivery**
  - Analog spectroscopic chain / Electrometer …

COSTS

VOICE	YEAR 1	YEAR 2
Missions	5k	5k
Disposables	10k	5k
SMU 4200A-SCS*	53k	
Pulsed Meas. U.		37k
DMCA + CSA		13k
SPC-Module		5k
TOTAL	68k	65k

*** Co-funding with Dip.Fis. Bologna**

Conclusions

PHASE-g aims at validating a ***novel approach*** to perovskite-based γ-ray detectors, which merges the bulk properties of Single Crystals with the major advantages of interdigitated structures.

It provides:

➡️ An Innovative dual structure which relies on **3D interdigitated electrodes**

➡️ A dedicated study on PMPS-based scintillating device for **photon counting**

➡️ A **spectroscopic device** for γ-rays of different doses and energies

➡️ Tests with **clinically relevant** radiation energies

It opens up new paradigms for detectors fabrication and thus scalability to **large-area**, **pixelated multielectrode arrays**.